Name: ______________________________

TOPIC TEST

Introduction to calculus (Advanced)

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 10 free-response questions (40 marks)
- Total: 50 marks

Part A

10 multiple-choice questions
1 mark each: 10 marks
Circle the correct answer.

1 $a = \frac{dv}{dt}$, where v represents velocity.

What does a not stand for?

A $\frac{\text{change in } v}{\text{change in } t}$

B instantaneous rate of change of velocity

C $f'(v)$

D acceleration

2 Which one of these functions is differentiable at $x = 1$?

A $y = (x + 1)(x - 1)$

B $y = \frac{1}{x-1}$

C $y = |x - 1|$

D $f(x) = \begin{cases} x^2 + 1, & x > 1 \\ x + 1, & x \le 1 \end{cases}$

3 Find the derivative of $f(x) = (2x^3 + 5)^4$.

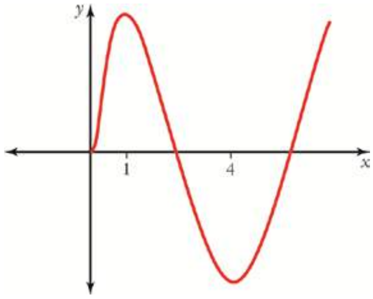
A $f'(x) = 24x(2x^3 + 5)^3$

B $f'(x) = 4x^2(2x^3 + 5)^3$

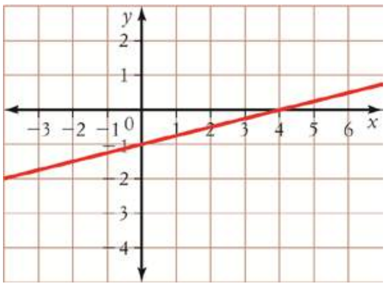
C $f'(x) = 24x^2(2x^3 + 5)^3$

D $f'(x) = 8x(2x^3 + 5)^3$

4 Which graph below is the gradient function of this graph?



A



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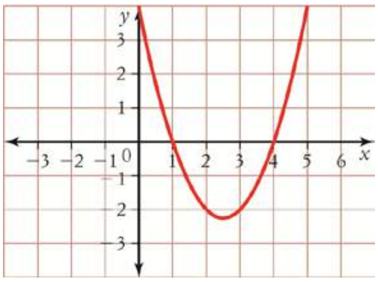

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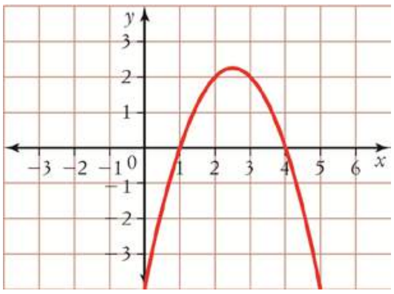

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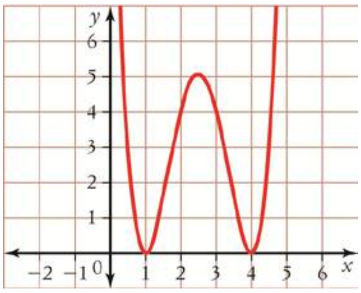


5 Find the difference quotient $\dfrac{f(x+h)-f(x)}{h}$ for $f(x) = x^2 - 4$.

A $-2x + h$ **B** $2x + h$

C $2xh$ **D** $2hx + h^2$

6 Find the value of $f'(-4)$ if $f(x) = 2x - x^3$.

A 56 **B** 50

C –50 **D** –46

7 Find the equation of the normal to the function $f(x) = 3x^2 - 2x + 1$ at $x = -1$.

A $6y - x - 49 = 0$ **B** $y - 8x - 49 = 0$

C $8y - x - 49 = 0$ **D** $y - 8x - 14 = 0$

8 Differentiate $y = \sqrt[3]{x^2}$.

A $\dfrac{dy}{dx} = \dfrac{2}{3\sqrt[3]{x}}$ **B** $\dfrac{dy}{dx} = \dfrac{2}{3\sqrt[3]{x}}$

C $\dfrac{dy}{dx} = 2\sqrt[3]{x}$ **D** $\dfrac{dy}{dx} = \dfrac{3}{2\sqrt[3]{x}}$

9 Differentiate $y = x(x^2 + 1)^4$.

A $y' = 8x^2(x^2 + 1)^3$

B $y' = 4x(x^2 + 1)^3 + (x^2 + 1)^4$

C $y' = 4x(x^2 + 1)^3$

D $y' = (x^2 + 1)^3(9x^2 + 1)$

10 A projectile is fired vertically into the air and its height in metres is given by $h = 40t - 5t^2 + 4$, where t is in seconds. Find when its velocity is zero.

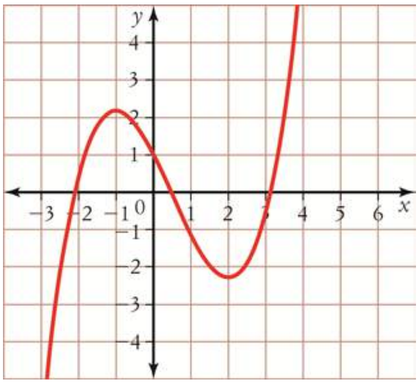
A 4 s **B** 5 s

C 7.5 s **D** 8 s

Part B

10 free-response questions
40 marks
Show your working where appropriate.

11 Draw on the same axes the gradient function of this graph.



[2 marks]

12 Differentiate each function from first principles.

a $f(x) = x^2 - 2x$

b $y = \frac{1}{x}$

[4 marks]

13 Find the values of x for which each function cannot be differentiated.

a $y = \dfrac{x}{x^2 - 1}$

b $y = x^2 + 2x^3 + 1$, where $x \neq 7$

c $y = |x + 2|$

[4 marks]

14 Find the derivative of each function after simplifying the function first.

a $f(x) = (x^2 - x)^2$

b $y = \dfrac{2x^5 - 6x^4 + 10x^3 + 2x^2}{2x^2}$

[4 marks]

15 Sketch $f(x) = x^3 - 8$ and $f'(x)$ on the same axes, labelling each graph with its equation and showing all intercepts.

[4 marks]

16 **a** Find the general form of the equation of the tangent to the hyperbola $y = \frac{5}{x}$ at $x = 2$.

b There is another tangent to the hyperbola that is parallel to the tangent found in part **a**. Find the coordinates of the point on the hyperbola at which this tangent exists.

[4 marks]

17 $f(x) = x^3 + 2x^2 - x + 1$ has a tangent at point $P(x, y)$ that is parallel to the graph of $2x + y = 7$.

a Find all the possible coordinates of P.

b Find the equation of the normal to the graph of $y = f(x)$ at $x = -1$.

[4 marks]

18 Differentiate each function.

a $y = (3x^2 + 1)(2x - 5)$

b $g(x) = \dfrac{x^3 - 1}{x + 2}$

c $h(x) = \sqrt{2x^4 - 2x}$

d $y = (2x^2 - 1)^2 (7x + 1)^3$

[8 marks]

19 $f(t) = t^2 - 12t + 2$.

a Find the average rate of change between $t = 1$ and $t = 4$.

b Find the instantaneous rate of change at $t = 4$.

[2 marks]

20 A ball is falling from the top of 350 m tower. The height of the ball is represented by $h = 350 + 5t - 5t^2$, where h is in metres and t is time in seconds.

a What is the height of the ball after 5 seconds?

b What is the average speed of the ball between 5 seconds and 10 seconds?

c What is the speed of the ball in the 10th second?

d At what speed would the ball hit the ground? Hence, what are the limitations of this model?

[4 marks]

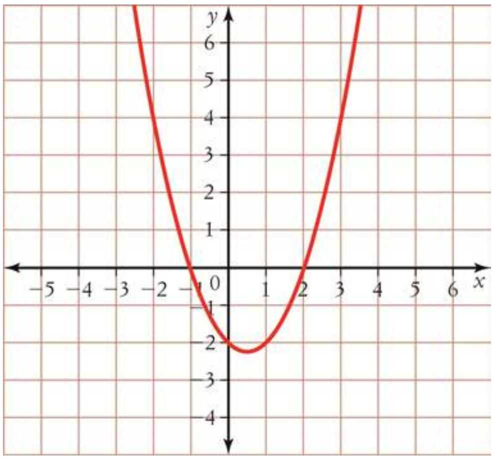
This is the end of the test.

Use the rest of this page for extra working space.

Answers

1 C **2** A **3** C **4** B **5** B
6 D **7** C **8** A **9** D **10** A

11

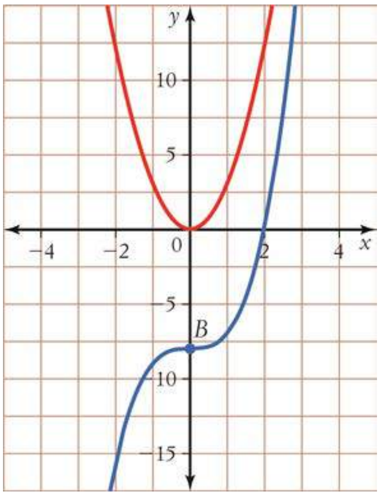


12 a $f'(x) = 2x - 2$ **b** $f'(x) = -\frac{1}{x^2}$

13 a $x = \pm 1$ **b** $x = 7$ **c** $x = -2$

14 a $f'(x) = 4x^3 - 6x^2 + 2x$ **b** $y' = 3x^2 - 6x + 5$

15



16 a $m = -\frac{5}{4}$, $5x + 4y - 20 = 0$ **b** $x = \pm 2$

17 a $\left(-\frac{1}{3}, \frac{41}{27}\right)$, $(-1, 3)$ **b** $2y - x - 7 = 0$

18 a $y' = 18x^2 - 30x + 2$ **b** $g'(x) = \frac{2x^3 + 6x^2 + 1}{(x+2)^2}$

c $\frac{4x^3 - 1}{\sqrt{2x^4 - 2x}}$ **d** $(7x + 1)^2(2x^2 - 1)(98x^2 + 8x - 21)$

19 a −7 **b** −4

20 a 250 m **b** −80 m/s **c** 95 m/s **d** 84 m/s, not valid after 8.9 seconds.